

SERA FREQUENCY UTILIZATION PLAN for 144 - 148 MHz	
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Some Definitions:

Narrow-band FM transmission is 2.5 kHz maximum deviation consuming 11 to 12.5 kHz of bandwidth

Wide-band FM transmission is 5 kHz maximum deviation consuming 13 to 16 kHz of bandwidth

Coordination parameter tags:

NBD - Narrow-band digital – DStar, NXDN, P25, DMR, 2.5kHz analog, 2.5kHz analog in mixed-mode

WBD - Wide-band digital – Fusion (must be coordinated on wide-band pair)

WBA - Wide-band analog – on designated wide-band pairs, 5.0kHz analog, 5.0kHz analog in mixed-mode

FM Voice Simplex Frequencies:

146.400*	146.415*	146.430*	146.445*	146.460*	146.475*
146.490*	146.505*	146.520	146.535	146.550	146.565
146.580	146.595	147.405*	147.420*	147.435*	147.450*
147.465*	147.480*	147.495*	147.510	147.525	147.540
147.555	147.570	147.585			

146.520 is the National FM Voice Simplex Calling Frequency

IMPORTANT NOTE: = Frequencies marked with * are permitted for simplex use as long as it does not cause interference to nearby repeater inputs. Please be aware that SERA coordinates some of these frequencies as repeater inputs in some areas. Check the Repeater Journal directory in your area to see if any such repeaters are present.

Repeater Sub-band 144.510 - 145.450 MHz: [NBD, WBD, WBA]

There are 20 repeater pairs (twenty 10 kHz channels) spaced at 20 kHz intervals using 600 kHz in/out split for wide-band and narrow-band transmission

144.510 - 145.110	144.530 - 145.130	144.550 - 145.150
144.570 - 145.170	144.590 - 145.190	144.610 - 145.210
144.630 - 145.230	144.650 - 145.250	144.670 - 145.270
144.690 - 145.290	144.710 - 145.310	144.730 - 145.330
144.750 - 145.350	144.770 - 145.370	144.790 - 145.390
144.810 - 145.410	144.830 - 145.430	144.850 - 145.450
144.870 - 145.470	144.890 - 145.490	

These pairs are available for the coordination and assignment of analog FM repeaters. Their coordination and assignment is determined using nominal FM coordination guidelines and other considerations, and those pertinent to co-channel and adjacent channel assignment.

19 Narrow-band Repeater Pairs 144.520 – 145.460: [NBD]

There are 19 repeater pairs (nineteen 10 kHz channels) spaced at 20 kHz intervals using 600 kHz in/out split for narrow-band transmission. DStar, DMR, NXDN, P25, NBFM, Mixed-Mode NBFM

144.520 - 145.120	144.540 - 145.140	144.560 - 145.160
144.580 - 145.180	144.600 - 145.200	144.620 - 145.220
144.640 - 145.240	144.660 - 145.260	144.680 - 145.280
144.700 - 145.300	144.720 - 145.320	144.740 - 145.340
144.760 - 145.360	144.780 - 145.380	144.800 - 145.400
144.820 - 145.420	144.840 - 145.440	144.860 - 145.460
144.880 - 145.480		

Wide Offset Digital Only Repeater Pairs: [NBD]

Dstar, DMR, NXDN, P25, NBFM, Mixed-Mode NBFM (System Fusion requires a wide-band channel).

4 Repeater Pairs, 20 kHz spacing, 2500 kHz in/out split

147.420 - 144.920	147.440 - 144.940	147.460 - 144.960	147.480 - 144.980
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4 Repeater Pairs, 20 kHz spacing, 1400 kHz in/out split

146.420 - 145.020	146.440 - 145.040	146.460 - 145.060	146.480 - 145.080
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IMPORTANT NOTE: Due to the prior assignment of frequencies in these ranges for alternate FM repeater inputs, these pairs may not initially be available in all areas.

Narrow-Band and Wide-Band Digital Packet Simplex Channels:

NARROW-BAND DIGITAL PACKET SIMPLEX CHANNELS: [NBD]

The following frequencies listed below are narrow-band digital simplex frequencies/channels and are located in the narrow-band section of the Amateur 2 meter band plan. They are interlaced in between narrow-band digital repeater outputs spaced 20 KHz apart. Standard FM deviation and modulation levels can exceed the 10 KHz channel bandwidth potentially impacting the reception performance of in-range repeater users. If any interference issues arise with a coordinated repeater, the coordinated repeater will have precedence over the digital simplex station. Consider using the wide-band digital simplex frequencies (listed next) before using a narrow-band digital simplex frequency.

144.910	144.930	144.950	144.970	144.990
145.010	145.030	145.050	145.070	145.090

Factors to consider when using these narrow-band digital simplex frequencies:

- Do not use wide-band FM VARA on these channels
- Ensure that *deviation is limited to 2.5 KHz* for all modes including FM packet stations
- Consider using an inexpensive SDR receiver (SDR-RTL) to evaluate and set bandwidth transmission limits
- Ask for help from knowledgeable and experienced Amateurs or technical persons including the Southeastern Repeater Association to avoid interfering with other stations.

WIDE-BAND DIGITAL PACKET SIMPLEX CHANNELS: [NBD, WBD]

The following Digital/Packet wide-band simplex frequencies are 20 KHz apart and suitable for wide-band FM VARA and FM Packet stations. Note that **145.510** and **145.790** are commonly published as DMR simplex frequencies. Note that the adjacent 145.800 frequency not listed below is reserved for ISS down-link communications.

145.510	145.530	145.550	145.570	145.590	145.610	145.630	145.650
145.670	145.690	145.710	145.730	145.750	145.770	145.790	

DStar - DMR - NXDN - P25 Digital Voice Simplex Frequencies: [NBD, WBD]

146.420*	146.440*	146.460*	146.480*	147.420*	147.440*
147.460*	147.480*				

IMPORTANT NOTE: = Frequencies marked with * are permitted for simplex use as long as it does not cause interference to nearby repeater inputs. Please be aware that SERA coordinates some of these frequencies as repeater inputs in some areas. Check the repeater directory in your area to see if any such repeaters are present. The use of these frequencies for D-Star operation, whether for simplex or repeater inputs, may impact FM simplex and repeater operations and their selection should be carefully considered.

Standard Repeater Band 146 - 148 MHz:

There are 68 repeater pairs (15 kHz channels) spaced at 15 kHz intervals using 600 kHz in/out split for wide-band and narrow-band transmission.

146.010 - 146.610	146.265 - 146.865	147.405 - 146.805	147.810 - 147.210
146.025 - 146.625	146.280 - 146.880	147.450 - 146.850*	147.855 - 147.255
146.040 - 146.640	146.295 - 146.895	147.495 - 146.895	147.900 - 147.300
146.055 - 146.655	146.310 - 146.910	147.420 - 146.820*	147.945 - 147.345
146.070 - 146.670	146.355 - 146.955	147.465 - 146.865*	147.990 - 147.390
146.085 - 146.685	146.325 - 146.925	147.435 - 146.835*	147.735 - 147.135
146.100 - 146.700	146.370 - 146.970	147.480 - 146.880*	147.780 - 147.180
146.115 - 146.715	146.340 - 146.940	147.600 - 147.000	147.825 - 147.225
146.130 - 146.730	146.385 - 146.985	147.645 - 147.045	147.870 - 147.270
146.145 - 146.745	146.400 - 147.000	147.690 - 147.090	147.915 - 147.315
146.160 - 146.760	146.445 - 147.045*	147.615 - 147.015	147.960 - 147.360
146.175 - 146.775	146.490 - 147.090*	147.660 - 147.060	147.750 - 147.150
146.190 - 146.790	146.415 - 147.015*	147.705 - 147.105	147.795 - 147.195
146.205 - 146.805	146.460 - 147.060*	147.630 - 147.030	147.840 - 147.240
146.220 - 146.820	146.505 - 147.105	147.675 - 147.075	147.885 - 147.285
146.235 - 146.835	146.430 - 147.030*	147.720 - 147.120	147.930 - 147.330
146.250 - 146.850	146.475 - 147.075*	147.765 - 147.165	147.975 - 147.375

IMPORTANT NOTE: = Due to the allocation of frequencies in this range for D-Star repeater inputs (see Wide Offset Digital-only Pairs on page 2), it is recommended that the frequency pairs marked with * using these alternate inputs not be selected for application and assignment for FM repeaters going forward.

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Frequency Spectrum Plan for 144 - 148 MHz

144.000 - 144.050 EME CW
144.050 - 144.100 General CW Operation
144.100 CW National Calling Frequency
144.100 - 144.200 EME and Weak Signal SSB
144.200 - 144.300 General SSB Operation
144.200 SSB National Calling Frequency
144.275 - 144.300 Propagation Beacons
144.300 - 144.500 Multi-Mode Operation
144.390 APRS Nationwide (Mode = 1200 Baud FM Packet)
144.510 - 144.890 FM and D-Star Repeater Inputs
144.910 - 145.090 FM Digital/Packet Simplex and D-Star Repeater Outputs
145.110 - 145.490 FM and D-Star Repeater Outputs
145.510 - 145.790 FM Digital/Packet Simplex (Auxiliary)
145.800 - 146.000 Satellite Sub-Band
146.010 - 146.385 FM and APCO P25 Repeater Inputs
146.400 - 146.585 FM Voice Simplex and Alternate Repeater Inputs
146.420 - 146.480 D-Star Voice Simplex and Repeater Inputs
146.520 FM National Calling Frequency
146.610 - 147.390 FM and APCO P25 Repeater Outputs
147.405 - 147.585 FM Voice Simplex and Alternate Repeater Inputs
147.420 - 147.480 D-Star Voice Simplex and Repeater Inputs
147.600 - 147.990 FM and APCO P25 Repeater Input

NOTE: Contact your respective SERA Director, Vice Director, or Frequency Coordinator, for information about the need for coordination of your system, before installing and operating the system. Most simplex voice and digital-packet operations are itinerant in nature, and as such will not be coordinated by SERA. Digital-Packet operators should contact their local digital-packet organization for additional information. Simplex cross band voice, as is possible with most over the counter dual band radios, is also considered itinerant in nature and will not be coordinated; however, users are asked to use the assigned frequencies shown in this document.

Version Updates:
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